CHAPTER 3

Emergent plants of permanent and seasonally-flooded wetlands

KEITH THOMPSON

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3.1 Introduction

Emergents typically emerge from water (such as a lake margin); floodplain plants may also do so, but only on a seasonal basis. When drainage is impeded, swamps develop, and these may be either permanent or seasonal (see Chapter 6). All aquatic and floodplain plants, therefore, grow on 'wetlands', and the I.U.C.N. definition given in Chapter 1 applies. However, because some floodplains

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are only inundated for perhaps two or three weeks in the year, it may be helpful to add a further definition of 'wetland' to those already offered:

> 'Wetlands' is a collective term for permanently or temporarily wet areas, shallow-water and land-water interfaces. The water may be fresh, brackish or saline, and wetlands are characterised, in their natural state, by plants and animals which are adapted to living in wet conditions (Environmental Council, 1984).

Thus, *Themeda triandra*, a common grass of the 'better' soils, some of which are seasonally flooded, is not especially adapted to flooding; *Leersia hexandra*, on the other hand, clearly is so, because its distribution is limited to the lower reaches of floodplains. All floodplain plants are emergents at one time or another (*i.e.* they are 'seasonal emergents') and, in order to be typical of floodplains, they must also be adapted in some way to being emergents, or to at least being very tolerant of waterlogging.

Unfortunately, although the African wetland and aquatic plant literature is quite extensive (see Chapter 10), it is almost exclusively regional and descriptive by nature. This chapter attempts to identify patterns on a continental basis and also to explain them in terms of autecology and synecology.

Although this book does not deal specifically with coastal or inland saline wetlands, such as coastal lagoons, saltmarshes, mangroves, dune lakes and slacks, seagrasses, salt pans, *etc.*, it is neither possible nor appropriate to ignore coastal and saline ecosystems altogether. The present chapter aims, wherever possible, not only to describe but also to explain the distribution of freshwater aquatic and semi-aquatic macrophytes. No such explanation can be valid unless the ecological extremes have been studied as well as the ecological norms. Much attention is paid in the following pages to seasonally flooded communities – the extreme case of plants tolerant of both flooding and of seasonal drought. It is equally important to examine the interface between saline and non-saline habitats, because the addition of osmotic stress to flooding/waterlogging tolerance imposes a very powerful selection pressure on species occurrence. It is as important to understand the selection pressures introduced by salinity as it is to appreciate that some semi-aquatic plants are restricted to sandy soils and others to base-rich substrates. Figure 3.13 shows how 'wetness' and salinity interact.

In contrast with temperate regions, there is no clear distinction in Africa between saline and non-saline soils. It is simply a matter of degree; tropical floodplains and seasonally wet soils are often saline, particularly on sand, because of the accumulation of evaporites. This chapter, therefore, ranges from the most salt-tolerant species, such as *Suada articulata, Diplachne fusca* and *Scirpus maritimus*, to aquatics such as *Miscanthidium* and *Scirpus cubensis* which are characteristic of high quality waters. It also covers the ground between halophytic shrubs such as *Tamarix, Pluchea* and *Pandanus* to large freshwater swamp emergents such as *Mitragyna* and *Syzygium*.

3.2 Distribution and diversity of species and communities

The diversity of wetland emergents is vastly greater than all euhydrophytes combined, and no attempt will be made here to reproduce a full list for the 30 million km^2 of the African continent. Rather, it is intended merely to characterise the different wetland types by discussing typical plant species and communities. The only literature of continent-wide value is Huntley and Walker (1982), Knapp (1973), Menaut (1983), Thompson and Hamilton (1983) and White (1983). The following regional references were also of particular value in the preparation of this Chapter: *Eastern and Central Africa* – Lind and Morrison (1974), Thompson (1976), Vesey-Fitzgerald (1963b, 1970); *Southern Africa* – Howard-Williams (1979), Mitchell (1978), Smith (1976), Thompson (1974); *Sudanian Zone* – Iltis and Lemoalle (1983), Lock (personal communication); *Guinea/Congo Zone* – Boughey (1957), Koechlin (1961), Lebrun and Gilbert (1954), Robyns (1947–1955), Sita (1970), Thompson (unpublished data), Van der Ben (1959). Additional literature is cross-indexed in Chapter 10.

It is quite instructive to look at some general statistics. Menaut (1983), for instance, notes that,